Exercise Set 16: Algebraic Topology

De Rham cohomology (Set 10) captures topological information via smooth differential forms. But there is a more fundamental approach: ***singular homology****, which counts cycles and boundaries using purely combinatorial data. The* ***fundamental group*** *$\pi_1(M)$ classifies loops up to homotopy and controls the covering space theory. Higher homotopy groups $\pi_n(M)$ classify spheres in M up to homotopy and encode the deepest topological structure. In physics, π_1 classifies topological defects (vortices, strings), π_2 classifies monopoles, and π_3 classifies instantons the integers that label these objects are exactly the homotopy groups.*

Recall of Definitions and Examples

Singular homology

R1. Write the definition of a **singular k-simplex** $\sigma : \Delta^k \to M$ where Δ^k is the standard k-simplex. Define the **chain group** $C_k(M;\mathbb{Z})$ and the **boundary map** $\partial : C_k \to C_{k-1}$. What is ∂^2?

R2. Define the **singular homology groups**: $H_k(M;\mathbb{Z}) = \ker\partial/\operatorname{im}\partial$. State the following:

(a) $H_0(M;\mathbb{Z}) = \mathbb{Z}^{\pi_0(M)}$ (one $\mathbb{Z}$ per connected component)

(b) $H_k(M;\mathbb{Z})$ for a contractible space

(c) The Euler characteristic: $\chi(M) = \sum(-1)^k \mathrm{rank}(H_k)$

R3. State the homology groups (from Nawata HW7):

(a) $H_k(S^n;\mathbb{Z})$

(b) $H_k(\Sigma_g;\mathbb{Z})$ for the Riemann surface of genus g

(c) $H_k(\mathbb{R}P^2;\mathbb{Z})$ note the torsion!

(d) $H_k(\text{Klein bottle};\mathbb{Z})$ more torsion

R4. What is **torsion** in homology? Give an example of a space with torsion in homology. Why does de Rham cohomology (which uses $\mathbb{R}$ coefficients) miss torsion?

Fundamental group and covering spaces

R5. Write the definition of $\pi_1(M,p)$: the group of loops based at $p \in M$ up to homotopy. What is the group operation? When is $\pi_1(M) = 0$ (simply connected)?

R6. State the fundamental groups (from Nawata HW9):

(a) $\pi_1(S^1)$

(b) $\pi_1(T^n)$

(c) $\pi_1(\Sigma_g)$ for $g \geq 1$

(d) $\pi_1(SO(3))$ and $\pi_1(SU(2))$ from Set 11

(e) $\pi_1(\mathbb{R}P^n)$ for $n \geq 2$

R7. Write the definition of a **covering space** $p : \tilde{M} \to M$. What is the **universal cover**? State the correspondence: connected covering spaces of $M \leftrightarrow$ subgroups of $\pi_1(M)$.

R8. Write the definitions of higher homotopy groups $\pi_n(M, p)$ for $n \geq 2$: homotopy classes of maps $S^n \to M$. State: π_n is abelian for $n \geq 2$. Give key examples:

(a) $\pi_n(S^n) = \mathbb{Z}$

(b) $\pi_n(S^m) = 0$ for $n < m$

(c) $\pi_3(S^2) = \mathbb{Z}$ (generated by the Hopf map)

(d) $\pi_n(SU(2)) = \pi_n(S^3)$

Theory

T1. (de Rham's theorem.)

State **de Rham's theorem**: integration gives an isomorphism:

$$H^k_{dR}(M; \mathbb{R}) \cong H^k(M; \mathbb{R}) = H^k(M; \mathbb{Z}) \otimes \mathbb{R}$$

via the pairing $(\omega, \sigma) \mapsto \int_\sigma \omega$.

(a) What does this say about the relationship between smooth analysis (de Rham) and combinatorial topology (singular)?

(b) Why does de Rham miss torsion?

(c) For $M = \Sigma_g$: verify the pairing between $H^1_{dR}(\Sigma_g)$ (dimension $2g$) and $H_1(\Sigma_g; \mathbb{Z}) = \mathbb{Z}^{2g}$.

T2. (The Hurewicz theorem.)

For a simply connected space ($\pi_1 = 0$) and $n \geq 2$: the **Hurewicz map** $h : \pi_n(M) \to H_n(M; \mathbb{Z})$ is an isomorphism if $\pi_k(M) = 0$ for all $k < n$.

(a) For $M = S^n$: verify $\pi_n(S^n) = H_n(S^n; \mathbb{Z}) = \mathbb{Z}$.

(b) The Hopf map $\eta : S^3 \to S^2$ generates $\pi_3(S^2) = \mathbb{Z}$ but $H_3(S^2) = 0$ the Hurewicz theorem does not apply because $\pi_2(S^2) = \mathbb{Z} \neq 0$. This shows π_n and H_n can differ.

T3. (Computing $\pi_1(\Sigma_g)$ Nawata HW9 P1.)

The Riemann surface Σ_g can be presented as a $4g$-gon with edges identified: $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b$

(a) Show $\pi_1(\Sigma_g)$ is generated by $a_1, b_1, \ldots, a_g, b_g$ with one relation: $[a_1, b_1][a_2, b_2] \cdots [a_g, b_g] = 1$ where $[a, b] = aba^{-1}b^{-1}$.

(b) For $g = 1$ (T^2): show $\pi_1(T^2) = \mathbb{Z}^2$ (abelian).

(c) For $g = 2$: $\pi_1(\Sigma_2)$ is non-abelian. Verify: the abelianization gives $H_1(\Sigma_2; \mathbb{Z}) = \mathbb{Z}^4$.

T4. (The universal cover of $SO(3)$.)

From Set 11: $\pi_1(SO(3)) = \mathbb{Z}_2$ with universal cover $SU(2)$.

(a) The covering map $p : SU(2) \to SO(3)$ has fiber $\mathbb{Z}_2 = \{I, -I\}$. Show: every path in $SO(3)$ from I to I lifts to either a loop in $SU(2)$ (if it lifts to a loop) or a path from I to $-I$.

(b) The loop in $SO(3)$ corresponding to a 360ř rotation lifts to a path from I to $-I$ in $SU(2)$ it is not a loop. The loop corresponding to 720ř lifts to a loop in $SU(2)$. This is the Dirac belt trick again.

(c) Conclude: $\pi_1(SO(3)) = \pi_0(\text{fiber}) = \mathbb{Z}_2$.

Computation

C1. (Homology of surfaces Nawata HW7 P3,4.)

(a) Compute $H_k(\Sigma_g; \mathbb{Z})$ using a CW structure: one 0-cell, $2g$ 1-cells, one 2-cell. Show: $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^{2g}$, $H_2 = \mathbb{Z}$.

(b) Compute $\chi(\Sigma_g) = 1 - 2g + 1 = 2 - 2g$. ✓

(c) For $\mathbb{R}P^2$: CW structure has one 0-cell, one 1-cell, one 2-cell with attaching map of degree 2. Compute $H_k(\mathbb{R}P^2; \mathbb{Z})$: $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}_2$, $H_2 = 0$.

(d) Identify the torsion $\mathbb{Z}_2 \subset H_1(\mathbb{R}P^2)$: the generator is the non-contractible loop. Why does $2 \cdot [\text{loop}] = 0$ in H_1?

C2. (The Hopf fibration and $\pi_3(S^2)$.)

The Hopf map $\eta : S^3 \to S^2$ is defined by: $\eta(z_0, z_1) = [z_0 : z_1] \in \mathbb{CP}^1 \cong S^2$.

(a) Show η is well-defined (the ratio $[z_0 : z_1]$ is invariant under the $U(1)$ action $(z_0, z_1) \mapsto (e^{i\theta} z_0, e^{i\theta} z_1)$).

(b) The fiber $\eta^{-1}(p)$ over any point $p \in S^2$ is a great circle $S^1 \subset S^3$. Show this for $p = [1 : 0]$.

(c) The Hopf map generates $\pi_3(S^2) = \mathbb{Z}$. The Hopf invariant of η is 1. For the map $\eta^n = n \cdot \eta$ (degree n in π_3): the Hopf invariant is n.

(d) **Physics**: the Hopf invariant classifies the winding of $SU(2) \cong S^3$ around S^2 it is the instanton number of the BPST instanton from Nawata HW11 when restricted to the boundary S^3.

C3. (Homotopy groups and topological defects.)

In a physical system with an order parameter taking values in a space M (the **order parameter space**), topological defects of codimension k are classified by $\pi_{k-1}(M)$.

(a) **Vortices** (codimension 2, i.e. line defects in 3D): classified by $\pi_1(M)$. For a superfluid with $M = U(1) = S^1$: $\pi_1(S^1) = \mathbb{Z}$ vortices are labelled by an integer (winding number). Write the winding number as an integral.

(b) **Monopoles** (codimension 3, point defects in 3D): classified by $\pi_2(M)$. For a ferromagnet with $M = S^2$ (spin direction): $\pi_2(S^2) = \mathbb{Z}$ monopoles are labelled by an integer. This is the Dirac monopole from Set 12.

(c) **Instantons** (codimension 4, point defects in 4D): classified by $\pi_3(M)$. For a Yang-Mills theory with $G = SU(2)$: $\pi_3(SU(2)) = \pi_3(S^3) = \mathbb{Z}$ instantons are labelled by an integer (the instanton number k from Nawata HW11).

(d) **Summary table**: fill in the homotopy group and physical interpretation for each:

Defect	Codim	Homotopy group	Example
Domain wall	1	$\pi_0(M)$	Ising model, $M = \{+, -\}$
Vortex	2	$\pi_1(M)$	Superfluid, $M = S^1$
Monopole	3	$\pi_2(M)$	Ferromagnet, $M = S^2$
Instanton	4	$\pi_3(M)$	YM, $M = SU(2) = S^3$

C4. (Lens spaces Nawata HW7 P5.)

The lens space $L(p, q)$ is constructed from n tetrahedra arranged cyclically (Nawata HW7 Problem 5).

(a) Compute the homology groups of $L(p, q)$. Show $H_1(L(p, q)) = \mathbb{Z}_p$.

(b) Show $\pi_1(L(p, q)) = \mathbb{Z}_p$.

(c) $L(7, 1)$ and $L(7, 2)$ have the same homology and fundamental group but are not homeomorphic (from our discussion). What invariant distinguishes them? *(Answer: the linking form a more refined invariant than H_*.)*

Physics: Topological Defects and Homotopy Theory

The classification of topological defects in condensed matter and field theory is entirely governed by homotopy groups. A defect is stable if the loop (or sphere) surrounding it represents a non-trivial element of the appropriate homotopy group. The integers

labelling these defects are topological invariants they cannot change under continuous deformations, only under phase transitions that change the order parameter space. This is homotopy theory applied directly to the real world.

P1. (Vortices in superfluids.)

A superfluid order parameter is $\Psi = |\Psi|e^{i\theta} \in \mathbb{C}$ with $|\Psi| = \text{const}$ far from defects, so the order parameter space is $M = U(1) = S^1$.

(a) Show $\pi_1(S^1) = \mathbb{Z}$. The generator is a loop that goes around once.

(b) A vortex of winding number n is a field configuration where θ winds n times as you go around the vortex: $\oint_\gamma d\theta = 2\pi n$. This is the winding number from Set 4.

(c) The superfluid velocity is $\mathbf{v} = \frac{\hbar}{m}\nabla\theta$. Show the circulation $\oint_\gamma \mathbf{v} \cdot d\ell = \frac{2\pi\hbar n}{m}$ is quantized. This is the Onsager-Feynman quantization of circulation.

(d) A vortex with $n = 1$ is stable you cannot continuously deform θ to remove the winding without passing through a singularity. This is the topological protection of the vortex.

P2. (Magnetic monopoles and π_2.)

A magnetic monopole in a gauge theory with gauge group G that is spontaneously broken to $H \subset G$ has order parameter space $M = G/H$. The monopoles are classified by $\pi_2(G/H)$.

(a) For $G = SU(2)$, $H = U(1)$ (the 't Hooft-Polyakov monopole): $G/H = S^2$, $\pi_2(S^2) = \mathbb{Z}$. Monopoles are labelled by an integer the Pontryagin number.

(b) The long exact sequence of the fibration $H \to G \to G/H$: $\cdots \to \pi_2(G) \to \pi_2(G/H) \to \pi_1(H) \to \pi_1(G) \to \cdots$ For $G = SU(2)$ (simply connected) and $H = U(1)$: $0 \to \mathbb{Z} \to \pi_1(U(1)) \to 0$, giving $\pi_2(S^2) = \mathbb{Z}$. ✓

(c) **Connection to Set 12**: the Dirac monopole is the abelian ($G = U(1)$) case. The 't Hooft-Polyakov monopole (Nawata HW11, Prasad-Sommerfield solution) is the non-abelian case. Both are classified by π_2.

P3. (Instantons and π_3.)

The Yang-Mills instanton is classified by $\pi_3(G)$. For $G = SU(2) \cong S^3$: $\pi_3(S^3) = \mathbb{Z}$.

(a) The instanton number (second Chern number) is: $k = \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge F) \in \mathbb{Z}$. This is the winding number of the map $g : S^3 \to SU(2)$ at the boundary of $\mathbb{R}^4$.

(b) From Nawata HW12: show $k = \deg(g)$ where $\deg(g)$ is the mapping degree of $g : S^3 \to S^3$.
Hint: $\frac{1}{24\pi^2}\int_{S^3} Tr(g^{-1}dg)^3 = \deg(g)$ *is the left-invariant volume form computation from Nawata HW12.*

(c) The BPST instanton ($k = 1$) is the generator of $\pi_3(SU(2))$. Higher instantons ($k > 1$) correspond to k widely separated BPST instantons.

(d) **Physical consequence**: instantons mediate quantum tunnelling between different topological vacua of the Yang-Mills theory. The theta vacuum $|\theta\rangle = \sum_n e^{in\theta}|n\rangle$ is a superposition of all topological sectors. The parameter θ (the QCD theta parameter) is physical its value is the deepest unsolved problem in the Standard Model (the strong CP problem).

P4. (The complete classification of defects.)

(a) A physical system with order parameter space M in d spatial dimensions can have defects of codimension k (dimension $d-k$) classified by $\pi_{k-1}(M)$. Fill in for a nematic liquid crystal with $M = \mathbb{R}P^2$:

- $\pi_0(\mathbb{R}P^2) = ? \to$ domain walls in 1D?
- $\pi_1(\mathbb{R}P^2) = ? \to$ vortices in 2D?
- $\pi_2(\mathbb{R}P^2) = ? \to$ monopoles in 3D?

(Answer: $\pi_1(\mathbb{R}P^2) = \mathbb{Z}_2$ there are $\mathbb{Z}_2$ vortices (disclinations) in liquid crystals.)

(b) The $\mathbb{Z}_2$ vortex is special: two vortices can annihilate (since $1 + 1 = 0$ in $\mathbb{Z}_2$) but one cannot. This is directly observable in liquid crystals.

(c) Write one sentence: why is the homotopy group classification of defects more fundamental than any specific field theory model?

The homotopy groups $\pi_n(M)$ classify maps $S^n \to M$ up to homotopy. In physics, they classify topological defects: vortices (π_1), monopoles (π_2), and instantons (π_3). The stability of a defect is a topological fact it cannot be understood from any local field theory argument.

This is topology protecting physics: the integers labelling vortices, monopoles, and instantons are invariants that no continuous deformation can change. They are as fundamental as the laws of motion themselves.***Reflection.*** *Algebraic topology assigns algebraic invariants to topological spaces: singular homology groups $H_k(M;\mathbb{Z})$ and homotopy groups $\pi_n(M)$. De Rham's theorem shows that de Rham cohomology and singular cohomology agree over $\mathbb{R}$, but singular homology captures more in particular the torsion that smooth forms cannot see.*

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